

Data Analysis & Visualization using Orange

A comprehensive hands-on practice manual designed to take you from beginner to confident data analyst. This workbook provides step-by-step guidance through Orange Data Mining Software, combining practical exercises with real-world applications. Whether you're a student starting your data science journey or an analyst expanding your toolkit, you'll gain practical skills in data workflow design, predictive modeling, clustering, and advanced analytics techniques.



Getting Started

How to Use This Workbook

01

Download Orange Software

Install Orange Data Mining Software on your computer. It's free, open-source, and works on Windows, Mac, and Linux systems.

03

Use Built-in Datasets

Orange includes sample datasets like Iris, Housing, and Titanic. Use these for practice unless otherwise specified.

02

Follow Sequential Modules

Work through each module in order. Each builds on previous concepts, creating a strong foundation for advanced techniques.

04

Save Your Workflows

After completing each module, save your workflow as an .ows file. This creates a portfolio of your learning journey.

Module 1

Building Your First Data Workflow

Understanding Orange's interface is the foundation of effective data analysis. The canvas is your workspace where widgets connect to form analytical workflows. Each widget performs a specific function, from loading data to creating visualizations. Think of widgets as LEGO blocks—individually simple, but powerful when combined.

Objective: Master the Orange interface, understand the canvas workspace, learn how widgets connect, and build your first complete analytical workflow.

Dataset: Iris (built-in) - a classic dataset containing measurements of 150 iris flowers across three species.

Quick Tip

The Iris dataset is perfect for beginners because it's small, clean, and clearly demonstrates classification concepts. You'll use it to understand data flow before tackling more complex datasets.

Module 1: Practice Tasks

1 Load Your Data

Drag the File widget onto the canvas and load the Iris dataset. This widget is your data entry point for any analysis.

2 View the Dataset

Connect the File widget to a Data Table widget. Click to see all 150 rows and 5 columns of iris measurements.

3 Analyze Distribution

Add a Distributions widget to visualize how values spread across each feature. Notice the bell curves and patterns.

4 Create Visualization

Connect to a Scatter Plot widget. Set Sepal Length on X-axis and Petal Length on Y-axis. See how species separate naturally.

5 Save Your Work

Click File → Save As and name your workflow "Module1_Iris.ows". You've built your first complete data pipeline!

Reflection Questions: What type of variables are present in the dataset (continuous or categorical)? Which feature best separates the three species visually?

Understanding Widgets and Data Channels

Objective: Master input-output connections and understand how data flows between widgets through channels.



Connect File → Select Columns → Data Table. Remove one feature (like Sepal Width) and observe how subsequent widgets adjust automatically.

Add a Box Plot widget and connect it to your data. Box plots show quartiles, revealing outliers and data spread visually.

Connect to Test & Score widget. Notice it requires both data AND a model input. This dual-input pattern is common in evaluation widgets.

Reflection: What error messages appear when channels are mismatched? Understanding the difference between data output and model output prevents workflow frustration.

Module 3

Hierarchical Clustering: Finding Natural Groups

Clustering discovers natural groupings in data without pre-labeled categories. Hierarchical clustering builds a tree (dendrogram) showing how data points merge into clusters at different similarity levels. This method is particularly useful when you don't know how many groups exist in your data beforehand.

Dataset: Iris - We'll see if clustering rediscovers the three species without using labels.

Step 1: Build Clusters

File → Hierarchical Clustering widget. The dendrogram appears immediately, showing merger patterns.

1

Step 3: Validate Results

Select clusters in the dendrogram, send to Data Table, and compare with actual species labels.

3

2

Step 2: Try Linkage Methods

Adjust linkage: Single (nearest points), Complete (farthest points), or Ward (minimize variance). Ward typically gives clearest groups.

Key Questions: Which linkage method produces the clearest separation? How many natural clusters appear before you reveal the labels? Does the algorithm find three groups matching the three iris species?

Module 4

Making Predictions with Machine Learning

Prediction is the core of supervised learning. You train a model on known examples, then it predicts outcomes for new data. Orange makes this process visual and intuitive—you can literally see your predictions flowing through the workflow.

Objective: Build predictive models and understand classification output.

Dataset: Iris - predict species from flower measurements.



1

Logistic Regression

A linear classifier that estimates probabilities. Simple but effective for well-separated classes.

2

Random Forest

Combines multiple decision trees. More complex but often more accurate, especially for nonlinear patterns.

3

View Predictions

Connect both models to Predictions widget. See predicted class vs. actual class side-by-side with probability scores.

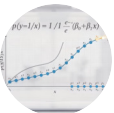
Analysis Questions: Which model gives better classification visually? What do the probability outputs tell you about model confidence? Notice cases where predictions differ—what makes those examples difficult?

Module 5: Model Evaluation & Scoring

Building models is only half the story—evaluating them properly is crucial. Test & Score widget provides a comprehensive evaluation framework, testing models on data they haven't seen during training. This reveals true predictive performance and helps you choose the best algorithm for your problem.



The Test & Score widget automatically performs cross-validation, splitting your data into training and testing sets multiple times to ensure robust results.



Logistic Regression

Fast, interpretable, works well for linearly separable data. Your baseline model.



Random Forest

Ensemble method combining multiple trees. Handles complex patterns and interactions.



k-Nearest Neighbors

Classifies based on proximity to training examples. Simple concept, powerful results.



Naïve Bayes

Probabilistic classifier based on Bayes' theorem. Efficient and surprisingly effective.

Interpretation: Compare Accuracy (overall correctness), AUC (ability to rank predictions), and F1 Score (balance of precision and recall). Random Forest often performs best due to its ability to capture complex feature interactions, but simpler models might be preferred for interpretability.

Module 6

Expanding Capabilities with Add-ons

Installation

Navigate to Options → Add-ons in the menu bar.
Browse available extensions and click Install for:

- Text Mining - analyze documents, tweets, reviews
- Image Analytics - classify and cluster images
- Network Analysis - visualize relationships
- Time Series - forecast temporal data

Orange's extensible architecture allows specialized widgets for domain-specific tasks. Add-ons transform Orange from a general data mining tool into a specialized platform for text analysis, image processing, network science, and more. Each add-on integrates seamlessly with core widgets, expanding your analytical toolkit without complexity.

Activity: After installation, explore the new widget categories that appear in your toolbar. Open example workflows included with each add-on to see professional applications. These extensions will be essential for upcoming modules on text and image analytics.

Module 7

Principal Components Analysis (PCA)

When datasets have many features, visualization and interpretation become challenging. PCA reduces dimensionality by creating new features (principal components) that capture maximum variance. The first few components often contain most of the information, allowing you to visualize high-dimensional data in 2D or 3D space.

01

Apply PCA

Connect File → PCA widget. The scree plot shows variance explained by each component.

02

Visualize Components

Send PCA output to Scatter Plot. Plot PC1 vs PC2 to see data in transformed space.

03

Interpret Results

Examine component loadings to understand what each principal component represents.



Key Questions: How much total variance is explained by PC1 alone? For Iris, PC1 typically captures 70-80% of variation. What happens when you select only 2 components—does clustering still work? Can you identify which original features contribute most to PC1 by examining the loadings?

Module 8

Feature Scoring & Ranking

Not all features are equally useful for prediction. Feature ranking identifies which variables provide the most information about your target. This insight guides feature selection—removing weak predictors can actually improve model performance by reducing noise and preventing overfitting.

Information Gain

Measures how much uncertainty about the target is reduced when you know the feature's value. Higher is better.

Gini Index

Used by decision trees, measures impurity reduction. Values near 0 indicate pure splits.

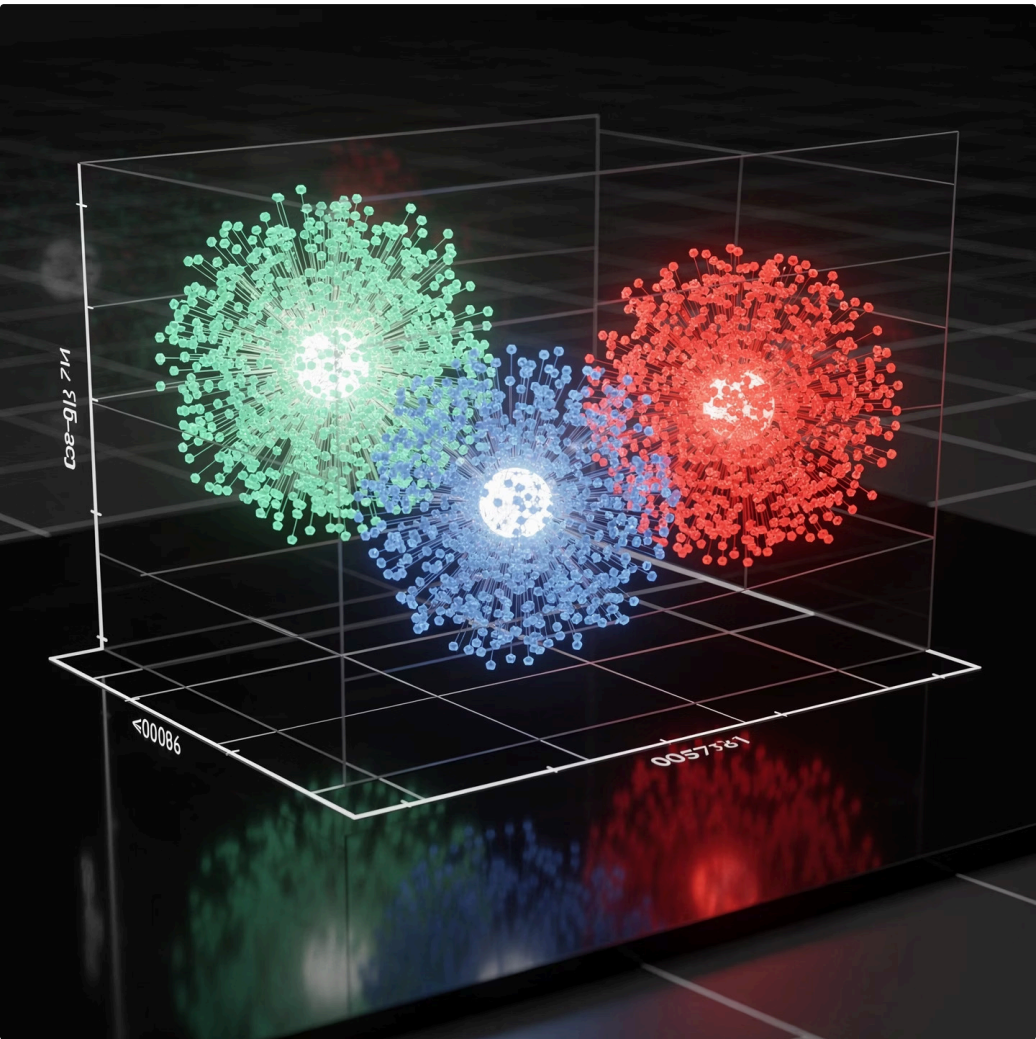
ANOVA

F-statistic tests if feature means differ significantly across classes. Higher F indicates stronger discrimination.

Practice Exercise: Use File → Rank widget and try all three scoring methods. Petal length and petal width typically rank highest for Iris classification. Now use Select Columns to remove the lowest-ranked feature, then re-run Test & Score. Does accuracy change? Often removing weak features maintains or even improves performance while simplifying the model.

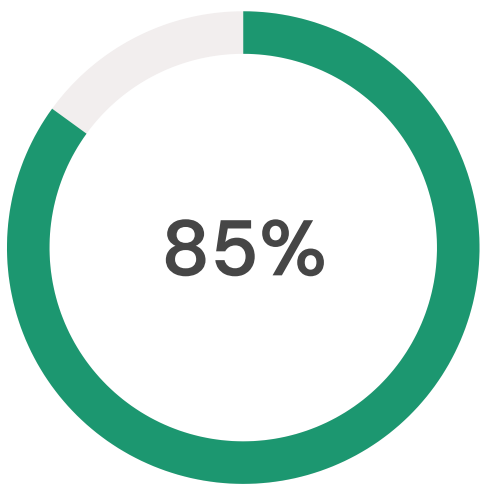
K-Means Clustering, Silhouette Score & MDS

K-Means is the most popular clustering algorithm, partitioning data into k distinct groups by minimizing within-cluster variance. Unlike hierarchical clustering, you must specify k in advance. The silhouette score measures cluster quality—values near +1 indicate well-separated clusters, while values near 0 suggest overlapping groups.



Practice Workflow:

- 1. File → k-Means widget, set k=3 (matching the three Iris species)
- 2. Connect to Silhouette Plot to assess cluster quality
- 3. Add MDS (Multidimensional Scaling) widget to visualize clusters in 2D space
- 4. Color points by cluster assignment vs. actual species



Typical match rate between k-means clusters and true Iris species labels



Average silhouette score indicating moderately separated clusters

Analysis Questions: What is your average silhouette score? Scores above 0.5 suggest reasonable clustering. Does the k-means grouping match actual species? One species usually clusters perfectly while the other two overlap slightly.

Module 10

Explaining K-Means Results

Finding clusters is just the beginning—understanding what makes each cluster unique is where insights emerge. By examining feature distributions within each cluster, you can characterize and name groups based on their distinguishing characteristics. This transforms abstract clusters into meaningful business segments or scientific categories.



Heat Map Analysis

Connect k-Means to Heat Map widget. Rows represent clusters, columns represent features. Color intensity shows feature values—quickly spot which features define each cluster.

Interpretation Exercise: What differentiates Cluster 1 from Cluster 2? For Iris, you might find: Cluster 1 has small petals (Setosa), Cluster 2 has medium measurements (Versicolor), Cluster 3 has large petals (Virginica). Write descriptions for each cluster based on feature patterns—this is how you translate math into meaning.



Box Plot Comparison

Add Box Plot widget showing feature distributions by cluster. Compare median values and spreads. One cluster might have large petals, another small—these patterns tell your cluster story.

Silhouette Image Analytics

Image analytics extends data mining to visual content. Orange's Image Analytics add-on uses deep learning embeddings to convert images into numerical features, enabling clustering and classification of photos without manual tagging. Silhouette analysis reveals how well images group together based on visual similarity.

Dataset: Use the sample image dataset or import your own collection of labeled images (minimum 20-30 images across 2-3 categories).



Load Images

Import Images widget loads your image folder. Preview thumbnails to verify import.

1

2

3

Cluster & Evaluate

Apply k-Means, then connect Silhouette Plot. High scores indicate visually similar images grouped together.

Create Embeddings

Image Embedding widget extracts features using pre-trained neural networks (SqueezeNet or Inception).

Key Question: How well are images clustered? Silhouette scores above 0.4 suggest the embedding captured meaningful visual patterns. Try different k values—optimal k often matches your true number of image categories.

Module 12

Image Classification with Neural Networks

Classification assigns images to predefined categories automatically. By combining deep learning embeddings with traditional machine learning algorithms, you can build accurate image classifiers without coding neural networks from scratch. This approach works well even with modest datasets (50-100 images per category).



Load & Embed

Import Images → Image Embedding. Neural network converts each image to a 1000+ dimension feature vector.



Train Classifier

Connect Logistic Regression (or Random Forest). Model learns to map embeddings to categories.



Evaluate

Use Test & Score to measure accuracy. Examine confusion matrix to see misclassification patterns.

Analysis Questions: What is your model's accuracy? Image classification typically achieves 70-90% accuracy depending on category similarity. Which class is misclassified most? Often categories with visual overlap (like "dog" vs. "wolf") confuse models. Use Image Viewer to examine misclassified examples—you'll often find ambiguous images even humans struggle with.

Introduction to Text Analytics

01

02

03

04

Bag of Words converts text to numerical matrix where rows are documents, columns are word counts.

[illegible]

Module 14

Text Clustering and Topic Modeling

Unsupervised text analysis discovers hidden themes without predefined categories. K-means clustering groups similar documents, while topic modeling extracts recurring word patterns that define themes. These techniques are invaluable for organizing large document collections and identifying emerging topics.

Document Clustering

Corpus → Bag of Words → k-Means. Start with $k=3-5$. Each cluster represents a theme or document type.

Topic Modeling

Use Topic Modeling widget (LDA algorithm). Specify number of topics. Each topic is a probability distribution over words.

Interpret Topics

Examine top words per topic. Name topics based on dominant terms (e.g., "technology", "politics", "sports").

Analysis Questions: How many topics appear meaningful? Too few topics are overly broad, too many are repetitive. For movie reviews, you might find topics like "acting quality", "plot summary", "technical production". What words define each cluster? Try 3-5 representative words per topic. Do clusters align with known document categories if available?

Text Classification: Sentiment & Category Prediction

Text classification assigns documents to categories automatically—sentiment analysis (positive/negative), spam detection, topic categorization, and more. By combining text preprocessing with supervised learning, you can build classifiers that understand document content and make predictions on new text.

Dataset: Use movie reviews or Twitter sentiment dataset (included with Text add-on).



The workflow follows standard supervised learning, but with text-specific preprocessing. Bag of Words creates thousands of features (one per unique word), so Logistic Regression's regularization prevents overfitting.

80%

Typical Accuracy

Sentiment classification on movie reviews

75%

Precision Score

Percentage of predicted positives that are correct

78%

Recall Score

Percentage of actual positives that are found

Interpretation: What is your accuracy? Examine precision vs. recall tradeoff—high precision means few false positives, high recall means few false negatives. For sentiment analysis, which type of error is worse: calling negative reviews positive or vice versa?

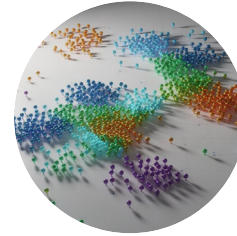
Modules 16–17

Advanced Text & Multivariate Techniques



Module 16: Import Documents

Use Corpus widget's "Import documents" option to load .txt files from folders. Each file becomes one document. Practice: Import 5 news articles on different topics, apply preprocessing and create word cloud to identify theme clusters automatically.



Module 17: Multivariate Projection

High-dimensional data needs dimension reduction for visualization. Compare techniques: MDS preserves distances between points, t-SNE reveals local structure and clusters, PCA maximizes variance. Each has strengths depending on your goal.

Projection Comparison Exercise:

1. Load Iris → connect to MDS, t-SNE, and PCA widgets
2. Send all outputs to Scatter Plot widgets
3. Compare visualizations side-by-side

Key Questions: Which projection separates classes best? t-SNE typically shows the clearest clusters because it emphasizes local neighborhood structure. Why is t-SNE better for visualization but PCA better for preprocessing? t-SNE is non-deterministic and doesn't preserve global structure, while PCA provides consistent, interpretable components.

Final Project & Certification Assignment

Apply everything you've learned to a real-world dataset. Choose either a customer churn prediction problem or marketing campaign analysis —both require comprehensive data mining workflows from preprocessing through interpretation.

1	Data Preprocessing Handle missing values, select features, normalize if needed
2	Feature Ranking Identify most important predictors using Rank widget
3	Classification Build and compare multiple models
4	Model Evaluation Use Test & Score to measure performance
5	Clustering Discover customer segments
6	Visualization Create PCA projection and interpretation plots
7	Business Insights Translate findings into recommendations

Submission Requirements

- Complete Orange workflow (.ows file)
- 2-page interpretation report
- Best model selection with justification
- Feature importance analysis
- Visualization screenshots
- Business insights and recommendations

Your report should tell a complete story: What patterns did you discover? Which customers are at risk of churning? What features predict campaign success? How confident are your predictions? What actions should the business take?

Remember: technical accuracy matters, but translating results into actionable insights is what makes data analysis valuable. Good luck!